

Example:

Given an AR(2) process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + z_t$$

Find MLE of σ_z^2 and approximate MLE of ϕ_1 and ϕ_2 .

Solution $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + z_t$

where $z_t \sim N(0, \sigma_z^2)$ is a purely random process. let $\{y_t; t=1, 2, \dots, n\}$ be the observed time series. We can define the likelihood function as,

$$L = f(y, \phi) = \prod_{t=3}^n f(y_t | y_{t-1}, y_{t-2})$$

$$L = f(y_1, y_2, \dots, y_n)$$

$$L = f(y_1, y_2) f(y_3 | y_1, y_2) f(y_4 | y_2, y_3) \dots f(y_n | y_{n-1}, y_{n-2})$$

$$\log L = \log f(y_1, y_2) + \sum_{t=3}^n \log f(y_t | y_{t-1}, y_{t-2}) \quad \text{--- (1)}$$

$$\underline{y} = (y_1, y_2) \sim N(\bar{\mu}, \Sigma_2)$$

$$f(y_1, y_2) = \frac{1}{(2\pi)^{1/2} |\Sigma_2|^{1/2}} e^{-\frac{1}{2} \underline{y}^T \Sigma_2^{-1} \underline{y}}$$

$$f(y_1, y_2) = \frac{1}{2\pi |\Sigma_2|^{1/2}} e^{-\frac{1}{2} \underline{y}^T \Sigma_2^{-1} \underline{y}}$$

$$\ln f(y_1, y_2) = \text{const} - \frac{1}{2} \ln |\Sigma_2| - \frac{1}{2} \underline{y}^T \Sigma_2^{-1} \underline{y}$$

$$\Sigma_2 = \sigma_z^2 \tilde{M}_2^{-1} \Rightarrow |\Sigma_2| = |\sigma_z^2 \tilde{M}_2^{-1}| = (\sigma_z^2)^2 |\tilde{M}_2^{-1}|$$

$$\ln f(\underline{y}_1, \underline{y}_2) = \text{Const} - \frac{1}{2} \ln[(\sigma_z^2)^2 |\underline{M}_2^{-1}|] - \frac{1}{2} \underline{y}^T (\sigma_z^2 \underline{M}_2^{-1}) \underline{y} \\ = \text{Const} - \frac{2}{2} \ln \sigma_z^2 - \frac{1}{2} \ln |\underline{M}_2^{-1}| - \frac{1}{2\sigma_z^2} (\underline{y}^T \underline{M}_2 \underline{y})$$

$$\ln f(\underline{y}_1, \underline{y}_2) = \text{Const} - \ln \sigma_z^2 + \frac{1}{2} \ln |\underline{M}_2| - \frac{1}{2\sigma_z^2} \underline{y}^T \underline{M}_2 \underline{y} \rightarrow (2)$$

$$\underline{y}_t = \phi_1 \underline{y}_{t-1} + \phi_2 \underline{y}_{t-2} + z_t +$$

$$E(\underline{y}_t | \underline{y}_{t-1}, \underline{y}_{t-2}) = \phi_1 \underline{y}_{t-1} + \phi_2 \underline{y}_{t-2} \quad \because E(z_t | \underline{y}_{t-1}, \underline{y}_{t-2}) = E(z_t) = 0$$

$$\text{Var}(\underline{y}_t | \underline{y}_{t-1}, \underline{y}_{t-2}) = \sigma_z^2$$

$$(\underline{y}_t | \underline{y}_{t-1}, \underline{y}_{t-2}) \sim N(\phi_1 \underline{y}_{t-1} + \phi_2 \underline{y}_{t-2}, \sigma_z^2)$$

$$f(\underline{y}_t | \underline{y}_{t-1}, \underline{y}_{t-2}) = \frac{1}{\sqrt{2\pi} \sigma_z} e^{-\frac{1}{2\sigma_z^2} [\underline{y}_t - \phi_1 \underline{y}_{t-1} - \phi_2 \underline{y}_{t-2}]^2}$$

$$\ln f(\underline{y}_t | \underline{y}_{t-1}, \underline{y}_{t-2}) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma_z^2 - \frac{1}{2\sigma_z^2} [\underline{y}_t - \phi_1 \underline{y}_{t-1} - \phi_2 \underline{y}_{t-2}]^2 \\ = \text{Const} - \frac{1}{2} \ln \sigma_z^2 - \frac{1}{2\sigma_z^2} [\underline{y}_t - \phi_1 \underline{y}_{t-1} - \phi_2 \underline{y}_{t-2}]^2$$

$$\sum_{t=3}^n \ln f(\underline{y}_t | \underline{y}_{t-1}, \underline{y}_{t-2}) = \text{Const} - \frac{(n-2)}{2} \ln \sigma_z^2 - \frac{1}{2\sigma_z^2} \sum_{t=3}^n [\underline{y}_t - \phi_1 \underline{y}_{t-1} - \phi_2 \underline{y}_{t-2}]^2 \rightarrow (3)$$

Put (2) and (3) in (1), we get

$$\ln L = \text{Const} - \ln \sigma_z^2 - \frac{1}{2} \ln |\underline{M}_2^{-1}| - \frac{1}{2\sigma_z^2} \underline{y}^T \underline{M}_2 \underline{y} \\ + \text{Const} - \frac{(n-2)}{2} \ln \sigma_z^2 - \frac{1}{2\sigma_z^2} \sum_{t=3}^n [\underline{y}_t - \phi_1 \underline{y}_{t-1} - \phi_2 \underline{y}_{t-2}]^2$$

$$\ln L = \text{Const} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln |\tilde{M}_2| - \frac{1}{2\sigma_z^2} \left[\sum_{t=3}^n (\tilde{y}_t - \phi_1 \tilde{y}_{t-1} - \phi_2 \tilde{y}_{t-2})^2 + \tilde{y}^T \tilde{M}_2 \tilde{y} \right] \rightarrow (4)$$

Diff. eq. (4) w.r.t. σ_z^2

$$\frac{\partial L}{\partial \sigma_z^2} = -\frac{n}{2\sigma_z^2} + \frac{1}{2\sigma_z^4} \left[\sum_{t=3}^n (\tilde{y}_t - \phi_1 \tilde{y}_{t-1} - \phi_2 \tilde{y}_{t-2})^2 + \tilde{y}^T \tilde{M}_2 \tilde{y} \right] = 0$$

$$-\frac{n}{0} + \frac{1}{\hat{\sigma}_z^2} \left[\sum_{t=3}^n (\tilde{y}_t - \phi_1 \tilde{y}_{t-1} - \phi_2 \tilde{y}_{t-2})^2 + \tilde{y}^T \tilde{M}_2 \tilde{y} \right] = 0$$

$$-n + \frac{1}{\hat{\sigma}_z^2} \left[\sum_{t=3}^n \tilde{z}_t^2 + \tilde{y}^T \tilde{M}_2 \tilde{y} \right] = 0$$

$$\frac{1}{\hat{\sigma}_z^2} = \frac{1}{n} \left[\sum_{t=3}^n \tilde{z}_t^2 + \tilde{y}^T \tilde{M}_2 \tilde{y} \right] \quad \checkmark \quad \text{Exact MLE}$$

For large 'n', we ignore $\tilde{y}^T \tilde{M}_2 \tilde{y}$

$$\frac{1}{\hat{\sigma}_z^2} = \frac{1}{n} \sum_{t=3}^n \tilde{z}_t^2 \quad \text{approximate MLE}$$

Partially differentiate (4) w.r.t. ϕ_1 and ϕ_2 . Assume 'n' is large and ignore $\frac{1}{2} \ln |\tilde{M}_2|$ and $\tilde{y}^T \tilde{M}_2 \tilde{y}$.

$$\frac{\partial \ln L}{\partial \phi_1} = 0 - 0 + 0 - \frac{2}{2\sigma_z^2} \left[\sum_{t=3}^n (\tilde{y}_t - \phi_1 \tilde{y}_{t-1} - \phi_2 \tilde{y}_{t-2}) (-\tilde{y}_{t-1}) \right] = 0$$

$$\frac{\cancel{\frac{1}{\sigma_z^2}}}{\sum_{t=3}^n \tilde{y}_{t-1}^2} \sum_{t=3}^n \tilde{y}_t \tilde{y}_{t-1} - \hat{\phi}_1 \frac{\sum_{t=3}^n \tilde{y}_{t-1}^2}{\sum_{t=3}^n \tilde{y}_{t-1}^2} - \phi_2 \frac{\sum_{t=3}^n \tilde{y}_{t-1} \tilde{y}_{t-2}}{\sum_{t=3}^n \tilde{y}_{t-1}^2} = 0$$

As n is large.

4

$$x_1 - \hat{\phi}_1 - \hat{\phi}_2 x_1 = 0$$

$$\hat{\phi}_1 = x_1(1 - \hat{\phi}_2) \rightarrow (5)$$

Similarly

$$\frac{\partial \ln L}{\partial \phi_2} = -\frac{1}{2\sigma^2} \left[2 \sum_{t=3}^n (y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2}) (-y_{t-2}) \right] = 0$$

$$\frac{\sum_{t=3}^n y_t y_{t-2} - \hat{\phi}_1 \sum_{t=3}^n y_{t-1} y_{t-2} - \hat{\phi}_2 \sum_{t=3}^n y_{t-2}^2}{\sum_{t=3}^n y_{t-2}^2} = 0$$

$$x_2 - \hat{\phi}_1 x_1 - \hat{\phi}_2 = 0$$

$$\hat{\phi}_2 = x_2 - \hat{\phi}_1 x_1 \rightarrow (6)$$

$$\hat{\phi}_2 = x_2 - x_1^2(1 - \hat{\phi}_2)$$

$$\hat{\phi}_2 = x_2 - x_1^2 + x_1^2 \hat{\phi}_2$$

$$\hat{\phi}_2 = \frac{x_2 - x_1^2}{1 - x_1^2} \rightarrow \text{approximate MLE}$$

$$\hat{\phi}_1 = x_1 [1 - x_2 + \hat{\phi}_1 x_1]$$

$$\hat{\phi}_1 = x_1 - x_1 x_2 + \hat{\phi}_1 x_1^2$$

$$\hat{\phi}_1 = \frac{x_1 - x_1 x_2}{1 - x_1^2} \rightarrow \text{approximate MLE}$$

Maximum likelihood estimation of AR(p) process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + z_t,$$

$z_t \sim N(0, \sigma_z^2)$ and $\{y_t; t=1, 2, \dots, n\}$ observed time series.

$$L = f(y_1, y_2, \dots, y_n) = f(y_1, y_2, \dots, y_p) f(y_{p+1} | y_1, y_2, \dots, y_p) \\ \dots f(y_n | y_{n-1}, \dots, y_{n-p})$$

$$L = f(y_1, y_2, \dots, y_p) \prod_{t=p+1}^n f(y_t | y_{t-1}, \dots, y_{t-p})$$

$$\ln L = \ln [f(y_1, y_2, \dots, y_p)] + \sum_{t=p+1}^n \ln [f(y_t | y_{t-1}, \dots, y_{t-p})]$$

$$\underline{y} = (y_1, \dots, y_p) \sim N_p(\underline{\mu}, \underline{\Sigma}_p)$$

$$f(y_1, \dots, y_p) = \frac{1}{(2\pi)^{p/2} |\underline{\Sigma}_p|^{1/2}} e^{-\frac{1}{2} \underline{y}^T \underline{\Sigma}_p^{-1} \underline{y}}$$

$$\ln f(y_1, \dots, y_p) = -\frac{p}{2} \ln(2\pi) - \frac{1}{2} \ln |\underline{\Sigma}_p| - \frac{1}{2} \underline{y}^T \underline{\Sigma}_p^{-1} \underline{y}$$

$$\text{As } \underline{\Sigma}_p = \sigma_z^2 \underline{M}_p^{-1}$$

$$\underline{\Sigma}_p^{-1} = \frac{1}{\sigma_z^2} \underline{M}_p$$

$$|\underline{\Sigma}_p| = |\sigma_z^2 \underline{M}_p^{-1}|$$

$$\underline{M}_p = \sigma_z^2 \underline{\Sigma}_p^{-1}$$

$$|\underline{\Sigma}_p| = (\sigma_z^2)^p |\underline{M}_p^{-1}|$$

$$\ln f(\underline{y}_1, \dots, \underline{y}_p) = -\frac{p}{2} \ln(2\pi) - \frac{p}{2} \ln \sigma_z^2 + \frac{1}{2} \ln |\underline{M}_p| - \frac{1}{2\sigma_z^2} \underline{y}^T \underline{M}_p \underline{y}$$

$$F(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p}) \sim N \left[\begin{matrix} E(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p}), \\ \text{Var}(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p}) \end{matrix} \right]$$

$$E(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p}) = \phi_1 \underline{y}_{t-1} + \dots + \phi_p \underline{y}_{t-p} = \mu_t^*$$

$$\text{Var}(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p}) = \sigma_z^2$$

$$f(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p}) = \frac{1}{\sqrt{2\pi} \sigma_z} e^{-\frac{1}{2\sigma_z^2} (\underline{y}_t - \mu_t^*)^2}$$

$$\ln f(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p}) = -\frac{1}{2} \ln(2\pi) - \frac{1}{2} \ln \sigma_z^2 - \frac{1}{2\sigma_z^2} (\underline{y}_t - \mu_t^*)^2$$

$$\sum_{t=p+1}^n \ln f(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p}) = -\frac{(n-p)}{2} \ln(2\pi) - \frac{(n-p)}{2} \ln \sigma_z^2 - \frac{1}{2\sigma_z^2} \sum_{t=p+1}^n (\underline{y}_t - \mu_t^*)^2$$

$$\ln L = \ln [f(\underline{y}_1, \dots, \underline{y}_p)] + \sum_{t=p+1}^n \ln [f(\underline{y}_t | \underline{y}_{t-1}, \dots, \underline{y}_{t-p})]$$

$$\ln L = -\frac{p}{2} \ln(2\pi) - \frac{p}{2} \ln \sigma_z^2 + \frac{1}{2} \ln |\underline{M}_p| - \frac{1}{2\sigma_z^2} \underline{y}^T \underline{M}_p \underline{y}$$

$$+ -\frac{(n-p)}{2} \ln(2\pi) - \frac{(n-p)}{2} \ln \sigma_z^2 - \frac{1}{2\sigma_z^2} \sum_{t=p+1}^n (\underline{y}_t - \mu_t^*)^2$$

$$\ln L = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln |\underline{M}_p| - \frac{1}{2\sigma_z^2} \left[\underline{y}^T \underline{M}_p \underline{y} + \sum_{t=p+1}^n (\underline{y}_t - \mu_t^*)^2 \right]$$

$$\ln L = \text{Const} - \frac{n}{2} \ln \sigma_z^2 + \frac{1}{2} \ln |\underline{M}_p| - \frac{1}{2\sigma_z^2} S(\phi_1, \phi_2, \dots, \phi_p),$$

$$\text{where } S(\phi_1, \phi_2, \dots, \phi_p) = \underline{y}^T \underline{M}_p \underline{y} + \sum_{t=p+1}^n (\underline{y}_t - \mu_t^*)^2$$

For MLE of σ_z^2

$$\frac{\partial \ln L}{\partial \sigma_z^2} = -\frac{n}{2\hat{\sigma}_z^2} + \frac{1}{2(\hat{\sigma}_z^2)^2} S(\phi_1, \dots, \phi_p) = 0$$

$$\hat{\sigma}_z^2 = \frac{S(\phi_1, \phi_2, \dots, \phi_p)}{n}$$

For MLE of $(\phi_1, \phi_2, \dots, \phi_p)$

$$\frac{\partial \ln L}{\partial \phi_j} = \frac{1}{2} \frac{\partial}{\partial \phi_j} \ln |M_p| - \frac{1}{2\sigma_z^2} \frac{\partial}{\partial \phi_j} S(\phi_1, \phi_2, \dots, \phi_p) = 0,$$

for $j = 1, 2, 3, \dots, p$

We cannot obtain precise MLEs of $\phi_1, \dots, \phi_p$ from above eqs because of challenging term $\frac{\partial}{\partial \phi_j} \ln |M_p|$. We can obtain approximate results by ignoring $\frac{\partial}{\partial \phi_j} \ln |M_p|$.

Least square estimation of AR(p) process

(1)

$$(y_t - \mu) = \phi_1(y_{t-1} - \mu) + \phi_2(y_{t-2} - \mu) + \dots + \phi_p(y_{t-p} - \mu) + z_t$$

Let $\{y_t; t=1, 2, \dots, n\}$ be the observed time series, we can write

$$z_t = (y_t - \mu) - \phi_1(y_{t-1} - \mu) - \phi_2(y_{t-2} - \mu) - \dots - \phi_p(y_{t-p} - \mu) \rightarrow$$

$$S = \sum_{t=p+1}^n z_t^2 = \sum_{t=p+1}^n [(y_t - \mu) - \phi_1(y_{t-1} - \mu) - \phi_2(y_{t-2} - \mu) - \dots - \phi_p(y_{t-p} - \mu)]^2 \rightarrow \textcircled{A}$$

$$\frac{\partial S}{\partial \mu} = 2 \sum_{t=p+1}^n [(y_t - \mu) - \phi_1(y_{t-1} - \mu) - \phi_2(y_{t-2} - \mu) - \dots - \phi_p(y_{t-p} - \mu)] \times [-1 + \phi_1 + \phi_2 + \dots + \phi_p] = 0$$

$$\sum_{t=p+1}^n [(y_t - \hat{\mu}) - \phi_1(y_{t-1} - \hat{\mu}) - \dots - \phi_p(y_{t-p} - \hat{\mu})] = 0$$

$$\sum_{t=p+1}^n (y_t - \hat{\mu}) - \phi_1 \sum_{t=p+1}^n (y_{t-1} - \hat{\mu}) - \dots - \phi_p \sum_{t=p+1}^n (y_{t-p} - \hat{\mu}) = 0$$

$$\sum_{t=p+1}^n y_t - (n-p)\hat{\mu} - \phi_1 \sum_{t=p+1}^n y_{t-1} + \phi_1(n-p)\hat{\mu} - \dots - \phi_p \sum_{t=p+1}^n y_{t-p} + \phi_p(n-p)\hat{\mu} = 0$$

$$\sum_{t=p+1}^n y_t - \phi_1 \sum_{t=p+1}^n y_{t-1} - \dots - \phi_p \sum_{t=p+1}^n y_{t-p} + (n-p)\hat{\mu}(1 - \phi_1 - \phi_2 - \dots - \phi_p)$$

$$\hat{\mu} = \frac{\sum_{t=p+1}^n y_t - \phi_1 \sum_{t=p+1}^n y_{t-1} - \dots - \phi_p \sum_{t=p+1}^n y_{t-p}}{(n-p)(1 - \phi_1 - \phi_2 - \dots - \phi_p)}$$

$$\hat{\mu} = \frac{1}{(1 - \phi_1 - \phi_2 - \dots - \phi_p)} \left[\frac{\sum_{t=p+1}^n y_t}{n-p} - \frac{\phi_1 \sum_{t=p+1}^n y_{t-1}}{n-p} - \dots - \frac{\phi_p \sum_{t=p+1}^n y_{t-p}}{n-p} \right]$$

For large n

$$\frac{\sum_{t=p+1}^n y_t}{n-p} \approx \frac{\sum_{t=p+1}^n y_{t-1}}{n-p} \approx \dots \approx \frac{\sum_{t=p+1}^n y_{t-p}}{n-p} \approx \bar{y}$$

$$\hat{\mu} = \frac{1}{(1 - \phi_1 - \dots - \phi_p)} (1 - \phi_1 - \dots - \phi_p) \bar{y}$$

$$\boxed{\hat{\mu} = \bar{y}}$$

$$t-1-(t-p) \\ t-1-t+p=p-1$$

To find values of the parameters $(\phi_1, \phi_2, \dots, \phi_p)$ we will differentiate eq. (A) w.r.t. $\phi_1, \phi_2, \dots, \phi_p$.

$$\frac{\partial S}{\partial \phi_1} = 2 \sum_{t=p+1}^n \left[(y_t - \hat{\mu}) - \hat{\phi}_1 (y_{t-1} - \hat{\mu}) - \dots - \hat{\phi}_p (y_{t-p} - \hat{\mu}) \right] \left[-(y_{t-1} - \hat{\mu}) \right] = 0$$

$$\sum_{t=p+1}^n (y_t - \hat{\mu})(y_{t-1} - \hat{\mu}) = \hat{\phi}_1 \sum_{t=p+1}^n (y_{t-1} - \hat{\mu})^2 + \dots + \hat{\phi}_p \sum_{t=p+1}^n (y_{t-p} - \hat{\mu})(y_{t-1} - \hat{\mu}) \quad \text{--- (1)}$$

Similarly we get

$$\sum_{t=p+1}^n (y_t - \hat{\mu})(y_{t-2} - \hat{\mu}) = \hat{\phi}_1 \sum_{t=p+1}^n (y_{t-1} - \hat{\mu})(y_{t-2} - \hat{\mu}) + \hat{\phi}_2 \sum_{t=p+1}^n (y_{t-2} - \hat{\mu})^2 + \dots + \sum_{t=p+1}^n (y_{t-p} - \hat{\mu})(y_{t-2} - \hat{\mu}) \quad \text{--- (2)}$$

$$\sum_{t=p+1}^n (y_t - \hat{\mu})(y_{t-p} - \hat{\mu}) = \hat{\phi}_1 \sum_{t=p+1}^n (y_{t-1} - \hat{\mu})(y_{t-p} - \hat{\mu}) + \dots + \sum_{t=p+1}^n (y_{t-p} - \hat{\mu})^2 \quad \text{--- (p)}$$

We can obtain the least square estimates of $\phi_1, \dots, \phi_p$ by solving 'p' eqs. simultaneously

Divide both sides of eq (1) by $\sum_{t=p+1}^n (y_{t-1} - \hat{\mu})^2$
 " " " " (2) " $\sum_{t=p+1}^n (y_{t-2} - \hat{\mu})^2$
 " " " " (p) " $\sum_{t=p+1}^n (y_{t-p} - \hat{\mu})^2$

We obtain approximate value of these equations

$$\begin{aligned}
 \hat{\phi}_1 + \hat{\phi}_2 \lambda_1 + \dots + \hat{\phi}_p \lambda_{p-1} &= \lambda_1 \\
 \hat{\phi}_1 \lambda_1 + \hat{\phi}_2 + \dots + \hat{\phi}_p \lambda_{p-2} &= \lambda_2 \\
 \hat{\phi}_1 \lambda_1 + \hat{\phi}_2 \lambda_2 + \dots + \hat{\phi}_p \lambda_{p-3} &= \lambda_3 \\
 \vdots &\vdots \\
 \hat{\phi}_1 \lambda_{p-1} + \hat{\phi}_2 \lambda_{p-2} + \dots + \hat{\phi}_p &= \lambda_p
 \end{aligned}
 \quad
 \begin{bmatrix}
 1 & \lambda_1 & \dots & \lambda_{p-1} \\
 \lambda_1 & 1 & \dots & \lambda_{p-2} \\
 \vdots & \vdots & \ddots & \vdots \\
 \lambda_{p-1} & \lambda_{p-2} & \dots & 1
 \end{bmatrix}
 \begin{bmatrix}
 \hat{\phi}_1 \\
 \hat{\phi}_2 \\
 \vdots \\
 \hat{\phi}_p
 \end{bmatrix}
 =
 \begin{bmatrix}
 \lambda_1 \\
 \lambda_2 \\
 \vdots \\
 \lambda_p
 \end{bmatrix}$$

$$\hat{\phi} R = \underline{\lambda} \Rightarrow \hat{\phi} = R^{-1} \underline{\lambda}$$

$(p \times 1) \quad (p \times p) \quad (p \times 1)$

which are Yule-Walker equations of AR(p) process. So, approx. LSEs of AR(p) process is solution of 'p' Yule-Walker equations.

Deductions

$p=1$; $\hat{\phi}_1 = \lambda_1$

$p=2$; $\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} 1 & \lambda_1 \\ \lambda_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}$

$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \frac{1}{1-\lambda_1^2} \begin{bmatrix} 1 & -\lambda_1 \\ -\lambda_1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}$

$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \frac{1}{1-\lambda_1^2} \begin{bmatrix} \lambda_1 - \lambda_1 \lambda_2 \\ -\lambda_1^2 + \lambda_2 \end{bmatrix}$

$\hat{\phi}_1 = \frac{\lambda_1 - \lambda_1 \lambda_2}{1 - \lambda_1^2}, \quad \hat{\phi}_2 = \frac{\lambda_2 - \lambda_1^2}{1 - \lambda_1^2}$

$\hat{\phi}_1 + \hat{\phi}_2 \lambda_1 = \lambda_1$
 $\hat{\phi}_1 \lambda_1 + \hat{\phi}_2 = \lambda_2$

$\begin{bmatrix} 1 & \lambda_1 \\ \lambda_1 & 1 \end{bmatrix} \begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix}$

2016

Find least square estimates of an AR(2) Process

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + Z_t$$

$$Z_t = Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2}$$

$$S = \sum_{t=3}^n Z_t^2 = \sum_{t=3}^n [Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2}]^2$$

Differentiate w.r.t. ϕ_1 and equate to zero

$$\frac{\partial S}{\partial \phi_1} = 2 \sum_{t=3}^n [Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2}] (-Y_{t-1}) = 0$$

$$\left[\sum_{t=3}^n Y_t - \hat{\phi}_1 \sum_{t=3}^n Y_{t-1} - \hat{\phi}_2 \sum_{t=3}^n Y_{t-2} \right] Y_{t-1} = 0$$

$$\sum_{t=3}^n Y_t Y_{t-1} - \hat{\phi}_1 \sum_{t=3}^n Y_{t-1}^2 - \hat{\phi}_2 \sum_{t=3}^n Y_{t-1} Y_{t-2} = 0$$

$$\hat{\phi}_1 \sum_{t=3}^n Y_{t-1}^2 + \hat{\phi}_2 \sum_{t=3}^n Y_{t-1} Y_{t-2} = \sum_{t=3}^n Y_t Y_{t-1} \quad \text{--- (1)}$$

Differentiate w.r.t. ϕ_2 and equate to zero

$$\frac{\partial S}{\partial \phi_2} = 2 \sum_{t=3}^n [Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2}] (-Y_{t-2}) = 0$$

$$\sum_{t=3}^n Y_t Y_{t-2} - \hat{\phi}_1 \sum_{t=3}^n Y_{t-1} Y_{t-2} - \hat{\phi}_2 \sum_{t=3}^n Y_{t-2}^2 = 0$$

$$\hat{\phi}_1 \sum_{t=3}^n Y_{t-1} Y_{t-2} + \hat{\phi}_2 \sum_{t=3}^n Y_{t-2}^2 = \sum_{t=3}^n Y_t Y_{t-2} \quad \text{--- (2)}$$

Divide both sides of eq. (1) and (2) by $(n-2)$, where (n) is sufficiently large (5)

$$\hat{\phi}_1 \frac{\sum_{t=3}^n y_{t-1}^2}{n-2} + \hat{\phi}_2 \frac{\sum_{t=3}^n y_{t-1} y_{t-2}}{n-2} = \frac{\sum_{t=3}^n y_t y_{t-1}}{n-2}$$

$$\hat{\phi}_1 \frac{\sum_{t=3}^n y_{t-1} y_{t-2}}{n-2} + \hat{\phi}_2 \frac{\sum_{t=3}^n y_{t-2}^2}{n-2} = \frac{\sum_{t=3}^n y_t y_{t-2}}{n-2}$$

According to stationary of $\{y_t\}_{t=1}^n$, we can write approximate version of above equations as

$$\hat{\phi}_1 C_0 + \hat{\phi}_2 C_1 = C_1$$

$$\hat{\phi}_1 C_1 + \hat{\phi}_2 C_0 = C_2$$

Divide both sides of above eqs. by C_0

$$\hat{\phi}_1 + \hat{\phi}_2 r_1 = r_1$$

$$\hat{\phi}_1 r_1 + \hat{\phi}_2 = r_2$$

$$\begin{bmatrix} 1 & r_1 \\ r_1 & 1 \end{bmatrix} \begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \begin{bmatrix} 1 & r_1 \\ r_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$= \frac{1}{1-r_1^2} \begin{bmatrix} 1 & -r_1 \\ -r_1 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\begin{bmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{bmatrix} = \frac{1}{1-r_1^2} \begin{bmatrix} r_1 - r_1 r_2 \\ -r_1^2 + r_2 \end{bmatrix}$$

$$\hat{\phi}_1 = \frac{r_1 - r_1 r_2}{1-r_1^2}$$

$$\hat{\phi}_2 = \frac{r_2 - r_1^2}{1-r_1^2}$$

Exp Derive least square estimate of an AR(1) process with zero mean

Sol $y_t = \phi_1 y_{t-1} + z_t$

$$z_t = y_t - \phi_1 y_{t-1}$$

Suppose $\{y_t; t=1, 2, \dots, n\}$ be the observed time series. The sum of squared errors of AR(1) process is defined as

$$S = \sum_{t=2}^n z_t^2 = \sum_{t=2}^n [y_t - \phi_1 y_{t-1}]^2$$

$$\frac{\partial S}{\partial \phi_1} = 2 \sum_{t=2}^n [y_t - \phi_1 y_{t-1}] (-y_{t-1}) = 0, \quad \frac{\partial^2 S}{\partial \phi_1^2} = 2 \sum_{t=2}^n (y_t - \phi_1 y_{t-1}) y_{t-1} > 0$$

$$\sum_{t=2}^n y_t y_{t-1} - \hat{\phi}_1 \sum_{t=2}^n y_{t-1}^2 = 0$$

$$\hat{\phi}_1 \sum_{t=2}^n y_{t-1}^2 = \sum_{t=2}^n y_t y_{t-1}$$

$$\hat{\phi}_1 = \frac{\sum_{t=2}^n y_t y_{t-1}}{\sum_{t=2}^n y_{t-1}^2}$$

$$\hat{\phi}_1 = \frac{C_1}{C_0} = \rho_1, \quad \boxed{\phi_1 = \rho_1}$$